

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product **Q**.

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .
- a_i is the amount of $\mathbf{Q}$ available at S_i .

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .
- a_i is the amount of $\mathbf{Q}$ available at S_i .
- d_j is the demand of $\mathbf{Q}$ at D_j .

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .
- a_i is the amount of $\mathbf{Q}$ available at S_i .
- d_j is the demand of $\mathbf{Q}$ at D_j .
- To find x_{ij} , $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, where x_{ij} is the amount of $\mathbf{Q}$ to be transported from S_i to D_j such that the demand at each D_j is met and the cost of transportation is minimum.

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .
- a_i is the amount of $\mathbf{Q}$ available at S_i .
- d_j is the demand of $\mathbf{Q}$ at D_j .
- To find x_{ij} , $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, where x_{ij} is the amount of $\mathbf{Q}$ to be transported from S_i to D_j such that the demand at each D_j is met and the cost of transportation is minimum.
- The problem is given by:
$$\text{Min } \sum_{i,j} c_{ij}x_{ij}$$
subject to

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .
- a_i is the amount of $\mathbf{Q}$ available at S_i .
- d_j is the demand of $\mathbf{Q}$ at D_j .
- To find x_{ij} , $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, where x_{ij} is the amount of $\mathbf{Q}$ to be transported from S_i to D_j such that the demand at each D_j is met and the cost of transportation is minimum.
- The problem is given by:

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$

subject to

$$\sum_{j=1}^n x_{ij} \leq a_i, \quad i = 1, 2, \dots, m$$

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .
- a_i is the amount of $\mathbf{Q}$ available at S_i .
- d_j is the demand of $\mathbf{Q}$ at D_j .
- To find x_{ij} , $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, where x_{ij} is the amount of $\mathbf{Q}$ to be transported from S_i to D_j such that the demand at each D_j is met and the cost of transportation is minimum.
- The problem is given by:

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$

subject to

$$\sum_{j=1}^n x_{ij} \leq a_i, i = 1, 2, \dots, m$$

$$\sum_{i=1}^m x_{ij} \geq d_j, j = 1, 2, \dots, n,$$

Transportation problem (TP)

- There are m supply stations $S_1, \dots, S_m$ for product $\mathbf{Q}$.
- There are n destination stations $D_1, D_2, \dots, D_n$ where $\mathbf{Q}$ is transported.
- c_{ij} is the cost of transportation of unit amount of $\mathbf{Q}$ from S_i to D_j .
- a_i is the amount of $\mathbf{Q}$ available at S_i .
- d_j is the demand of $\mathbf{Q}$ at D_j .
- To find x_{ij} , $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$, where x_{ij} is the amount of $\mathbf{Q}$ to be transported from S_i to D_j such that the demand at each D_j is met and the cost of transportation is minimum.
- The problem is given by:

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$

subject to

$$\sum_{j=1}^n x_{ij} \leq a_i, \quad i = 1, 2, \dots, m$$

$$\sum_{i=1}^m x_{ij} \geq d_j, \quad j = 1, 2, \dots, n,$$

$$x_{ij} \geq 0 \text{ for } i = 1, 2, \dots, m, j = 1, 2, \dots, n.$$

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.
- In that case all the inequalities in the constraints should hold as equalities.

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.
- In that case all the inequalities in the constraints should hold as equalities.
- A balanced transportation problem (TP) is given by,
 $\text{Min } \sum_{i,j} c_{ij}x_{ij}$
 subject to

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.
- In that case all the inequalities in the constraints should hold as equalities.
- A balanced transportation problem (TP) is given by,

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$
 subject to

$$\sum_{j=1}^n x_{ij} = a_i, \quad i = 1, 2, \dots, m$$

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.
- In that case all the inequalities in the constraints should hold as equalities.
- A balanced transportation problem (TP) is given by,

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$
 subject to

$$\sum_{j=1}^n x_{ij} = a_i, i = 1, 2, \dots, m$$

$$\sum_{i=1}^m x_{ij} = d_j, j = 1, 2, \dots, n,$$

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.
- In that case all the inequalities in the constraints should hold as equalities.
- A balanced transportation problem (TP) is given by,

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$
 subject to

$$\sum_{j=1}^n x_{ij} = a_i, i = 1, 2, \dots, m$$

$$\sum_{i=1}^m x_{ij} = d_j, j = 1, 2, \dots, n,$$

$$x_{ij} \geq 0 \text{ for } i = 1, 2, \dots, m, j = 1, 2, \dots, n.$$

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.
- In that case all the inequalities in the constraints should hold as equalities.
- A balanced transportation problem (TP) is given by,

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$
 subject to

$$\sum_{j=1}^n x_{ij} = a_i, i = 1, 2, \dots, m$$

$$\sum_{i=1}^m x_{ij} = d_j, j = 1, 2, \dots, n,$$

$$x_{ij} \geq 0 \text{ for } i = 1, 2, \dots, m, j = 1, 2, \dots, n.$$
- Since $\sum_i a_i = \sum_j d_j$

- It is clear that for the transportation problem (TP) to be **feasible** $\sum_i a_i \geq \sum_j d_j$.
- A transportation problem (TP) is called **balanced** if $\sum_i a_i = \sum_j d_j$.
- In that case all the inequalities in the constraints should hold as equalities.
- A balanced transportation problem (TP) is given by,

$$\text{Min } \sum_{i,j} c_{ij} x_{ij}$$
 subject to

$$\sum_{j=1}^n x_{ij} = a_i, i = 1, 2, \dots, m$$

$$\sum_{i=1}^m x_{ij} = d_j, j = 1, 2, \dots, n,$$

$$x_{ij} \geq 0 \text{ for } i = 1, 2, \dots, m, j = 1, 2, \dots, n.$$
- Since $\sum_i a_i = \sum_j d_j$
 if $\mathbf{x} = (x_{ij})_{m \times n}$ satisfies any $(m + n - 1)$ equations then it automatically satisfies all the $(m + n)$ equations.

- The constraints are of the form

$$Ax = b,$$

where

$$A_{(m+n) \times mn} = \begin{bmatrix} \overbrace{111\dots 11}^n & \mathbf{0}_n & \mathbf{0}_n & \dots & \cdot & \mathbf{0}_n \\ \mathbf{0}_n & \overbrace{111\dots 11}^n & \mathbf{0}_n & \dots & \cdot & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{0}_n & \overbrace{111\dots 11}^n & \mathbf{0}_n & \cdot & \mathbf{0}_n \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \mathbf{0}_n & \cdot & \cdot & \dots & \mathbf{0}_n & \overbrace{111\dots 11}^n \\ \overbrace{100\dots 0} & \overbrace{100\dots 0} & \cdot & \cdot & \dots & \overbrace{100\dots 0} \\ \overbrace{010\dots 0} & \overbrace{010\dots 0} & \cdot & \cdot & \dots & \overbrace{010\dots 0} \\ \cdot & \cdot & \cdot & \dots & \dots & \cdot \\ \overbrace{000\dots 01} & \overbrace{000\dots 01} & \cdot & \cdot & \dots & \overbrace{000\dots 01} \end{bmatrix}$$

- The constraints are of the form

$$A\mathbf{x} = \mathbf{b},$$

where

$$A_{(m+n) \times mn} = \begin{bmatrix} \overbrace{111\dots 11}^n & \mathbf{0}_n & \mathbf{0}_n & \dots & \cdot & \mathbf{0}_n \\ \mathbf{0}_n & \overbrace{111\dots 11}^n & \mathbf{0}_n & \dots & \cdot & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{0}_n & \overbrace{111\dots 11}^n & \mathbf{0}_n & \cdot & \mathbf{0}_n \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \mathbf{0}_n & \cdot & \cdot & \dots & \mathbf{0}_n & \overbrace{111\dots 11}^n \\ \overbrace{100\dots 0} & \overbrace{100\dots 0} & \cdot & \cdot & \dots & \overbrace{100\dots 0} \\ \overbrace{010\dots 0} & \overbrace{010\dots 0} & \cdot & \cdot & \dots & \overbrace{010\dots 0} \\ \cdot & \cdot & \cdot & \dots & \dots & \cdot \\ \overbrace{000\dots 01} & \overbrace{000\dots 01} & \cdot & \cdot & \dots & \overbrace{000\dots 01} \end{bmatrix}$$

$\mathbf{b} = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_n]^T$ (the $\mathbf{0}_n$'s are row vectors with n components).

- The constraints are of the form

$$Ax = b,$$

where

$$A_{(m+n) \times mn} = \begin{bmatrix} \overbrace{111\dots 11}^n & \mathbf{0}_n & \mathbf{0}_n & \dots & \cdot & \mathbf{0}_n \\ \mathbf{0}_n & \overbrace{111\dots 11}^n & \mathbf{0}_n & \dots & \cdot & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{0}_n & \overbrace{111\dots 11}^n & \mathbf{0}_n & \cdot & \mathbf{0}_n \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \mathbf{0}_n & \cdot & \cdot & \dots & \mathbf{0}_n & \overbrace{111\dots 11}^n \\ \overbrace{100\dots 0} & \overbrace{100\dots 0} & \cdot & \cdot & \dots & \overbrace{100\dots 0} \\ \overbrace{010\dots 0} & \overbrace{010\dots 0} & \cdot & \cdot & \dots & \overbrace{010\dots 0} \\ \cdot & \cdot & \cdot & \dots & \dots & \cdot \\ \overbrace{000\dots 01} & \overbrace{000\dots 01} & \cdot & \cdot & \dots & \overbrace{000\dots 01} \end{bmatrix}$$

$\mathbf{b} = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_n]^T$ (the $\mathbf{0}_n$'s are row vectors with n components).

- $\text{rank}(A) = m + n - 1$.

- Remove the last equation (any equation) from the $(m + n)$ equations.

- Remove the last equation (any equation) from the $(m + n)$ equations.
- In the new $\mathbf{Ax} = \mathbf{b}$, the dimension of A is $(m + n - 1) \times mn$ and $\mathbf{b} = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_{n-1}]^T$.

- Remove the last equation (any equation) from the $(m + n)$ equations.
- In the new $\mathbf{Ax} = \mathbf{b}$, the dimension of A is $(m + n - 1) \times mn$ and $\mathbf{b} = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_{n-1}]^T$.
- Any BFS of this problem will have $m + n - 1$ **basic variables** and the order of any **basis matrix**, $\mathbf{B}$ is $(m + n - 1) \times (m + n - 1)$.

- Remove the last equation (any equation) from the $(m + n)$ equations.
- In the new $\mathbf{Ax} = \mathbf{b}$, the dimension of A is $(m + n - 1) \times mn$ and $\mathbf{b} = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_{n-1}]^T$.
- Any BFS of this problem will have $m + n - 1$ **basic variables** and the order of any **basis matrix**, $\mathbf{B}$ is $(m + n - 1) \times (m + n - 1)$.
- **Theorem 1** : Let $\mathbf{B}$ be a basis matrix then:
 1. There exists a **row** of $\mathbf{B}$ with **exactly one** nonzero entry (which is a 1).

- Remove the last equation (any equation) from the $(m + n)$ equations.
- In the new $\mathbf{Ax} = \mathbf{b}$, the dimension of A is $(m + n - 1) \times mn$ and $\mathbf{b} = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_{n-1}]^T$.
- Any BFS of this problem will have $m + n - 1$ **basic variables** and the order of any **basis matrix**, $\mathbf{B}$ is $(m + n - 1) \times (m + n - 1)$.
- **Theorem 1** : Let $\mathbf{B}$ be a basis matrix then:
 1. There exists a **row** of $\mathbf{B}$ with **exactly one** nonzero entry (which is a 1).
 2. The sub matrix obtained by **deleting** the corresponding **row and column** (containing that nonzero entry) from $\mathbf{B}$ will again be **nonsingular** and will have a row with a **single nonzero** entry.

- Remove the last equation (any equation) from the $(m + n)$ equations.
- In the new $\mathbf{Ax} = \mathbf{b}$, the dimension of A is $(m + n - 1) \times mn$ and $\mathbf{b} = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_{n-1}]^T$.
- Any BFS of this problem will have $m + n - 1$ **basic variables** and the order of any **basis matrix**, $\mathbf{B}$ is $(m + n - 1) \times (m + n - 1)$.
- **Theorem 1** : Let $\mathbf{B}$ be a basis matrix then:
 1. There exists a **row** of $\mathbf{B}$ with **exactly one** nonzero entry (which is a 1).
 2. The sub matrix obtained by **deleting** the corresponding **row and column** (containing that nonzero entry) from $\mathbf{B}$ will again be **nonsingular** and will have a row with a **single nonzero** entry.
- If $\mathbf{B}$ is a **square sub matrix** of A with properties 1 and 2 of theorem 1, then $|\mathbf{B}| = \pm 1$

- Remove the last equation (any equation) from the $(m + n)$ equations.
- In the new $Ax = b$, the dimension of A is $(m + n - 1) \times mn$ and $b = [a_1, a_2, \dots, a_m, d_1, d_2, \dots, d_{n-1}]^T$.
- Any BFS of this problem will have $m + n - 1$ **basic variables** and the order of any **basis matrix**, B is $(m + n - 1) \times (m + n - 1)$.
- **Theorem 1** : Let B be a basis matrix then:
 1. There exists a **row** of B with **exactly one** nonzero entry (which is a 1).
 2. The sub matrix obtained by **deleting** the corresponding **row and column** (containing that nonzero entry) from B will again be **nonsingular** and will have a row with a **single nonzero** entry.
- If B is a **square sub matrix** of A with properties 1 and 2 of theorem 1, then $|B| = \pm 1$
- D is any **nonsingular** sub matrix of A if and only if D has the same structure as B .

- If D is any square submatrix of A , then $|D|$ is either $0, 1$ or -1 .

- If D is any square submatrix of A , then $|D|$ is either $0, 1$ or -1 .
- Such matrices (such as B) are called triangular matrices, and because of this special structure of B it is easy to solve system of equations of the form $Bx_B = b$ (which will give a basic solution of the transportation problem).

- If D is any square submatrix of A , then $|D|$ is either $0, 1$ or -1 .
- Such matrices (such as B) are called triangular matrices, and because of this special structure of B it is easy to solve system of equations of the form $Bx_B = b$ (which will give a basic solution of the transportation problem).
- If the i th row of B has a single nonzero entry at the j th column, then assign $x_{ij} = b_i$ (where b_i is either a_i or d_j).

- If D is any square submatrix of A , then $|D|$ is either $0, 1$ or -1 .
- Such matrices (such as B) are called triangular matrices, and because of this special structure of B it is easy to solve system of equations of the form $Bx_B = b$ (which will give a basic solution of the transportation problem).
- If the i th row of B has a single nonzero entry at the j th column, then assign $x_{ij} = b_i$ (where b_i is either a_i or d_j).
- Remove the i th row and the j th column from B which gives the matrix B_1 .

- If D is any square submatrix of A , then $|D|$ is either $0, 1$ or -1 .
- Such matrices (such as B) are called triangular matrices, and because of this special structure of B it is easy to solve system of equations of the form $Bx_B = b$ (which will give a basic solution of the transportation problem).
- If the i th row of B has a single nonzero entry at the j th column, then assign $x_{ij} = b_i$ (where b_i is either a_i or d_j).
- Remove the i th row and the j th column from B which gives the matrix B_1 .
- Solve the system $B_1 x' = b'$,

- If D is any square submatrix of A , then $|D|$ is either $0, 1$ or -1 .
- Such matrices (such as B) are called triangular matrices, and because of this special structure of B it is easy to solve system of equations of the form $Bx_B = b$ (which will give a basic solution of the transportation problem).
- If the i th row of B has a single nonzero entry at the j th column, then assign $x_{ij} = b_i$ (where b_i is either a_i or d_j).
- Remove the i th row and the j th column from B which gives the matrix B_1 .
- Solve the system $B_1x' = b'$, where x' is obtained from x by removing x_{ij} and b' is obtained from b by removing b_i and changing b_j to $b_j - b_i$.

- If D is any square submatrix of A , then $|D|$ is either $0, 1$ or -1 .
- Such matrices (such as B) are called triangular matrices, and because of this special structure of B it is easy to solve system of equations of the form $Bx_B = b$ (which will give a basic solution of the transportation problem).
- If the i th row of B has a single nonzero entry at the j th column, then assign $x_{ij} = b_i$ (where b_i is either a_i or d_j).
- Remove the i th row and the j th column from B which gives the matrix B_1 .
- Solve the system $B_1x' = b'$, where x' is obtained from x by removing x_{ij} and b' is obtained from b by removing b_i and changing b_j to $b_j - b_i$.
- Continuing in this way solve the system $Bx = b$.

- In any **basic solution** the **basic variables** takes values of the form, $\sum_i \gamma_i b_i$, where the γ_i 's are either **0,1** or **-1**.

- In any **basic solution** the **basic variables** takes values of the form, $\sum_i \gamma_i b_i$, where the γ_i 's are either **0,1** or **-1**.
- In any **BFS x** of **TP** with supplies $a_i, i = 1, 2, \dots, m$ and demand $d_j, j = 1, 2, \dots, n$ the basic variables take values of the form,

- In any **basic solution** the **basic variables** takes values of the form, $\sum_i \gamma_i b_i$, where the γ_i 's are either **0,1** or **-1**.
- In any **BFS x** of **TP** with supplies a_i , $i = 1, 2, \dots, m$ and demand d_j , $j = 1, 2, \dots, n$ the basic variables take values of the form,

$$x_{ij} = \sum_i \alpha_i a_i + \sum_j \beta_j d_j$$

- In any **basic solution** the **basic variables** takes values of the form, $\sum_i \gamma_i b_i$, where the γ_i 's are either **0,1** or **-1**.
- In any **BFS x** of **TP** with supplies a_i , $i = 1, 2, \dots, m$ and demand d_j , $j = 1, 2, \dots, n$ the basic variables take values of the form,
 $x_{ij} = \sum_i \alpha_i a_i + \sum_j \beta_j d_j$ where the α_i 's, β_j 's are either **0,1** or **-1**.

- In any **basic solution** the **basic variables** takes values of the form, $\sum_i \gamma_i b_i$, where the γ_i 's are either **0,1** or **-1**.
- In any **BFS x** of **TP** with supplies a_i , $i = 1, 2, \dots, m$ and demand d_j , $j = 1, 2, \dots, n$ the basic variables take values of the form,
 $x_{ij} = \sum_i \alpha_i a_i + \sum_j \beta_j d_j$ where the α_i 's, β_j 's are either **0,1** or **-1**.
- **Transportation Array (TA):** The mn variables x_{ij} can be arranged in an $m \times n$ array known as the $m \times n$ **transportation array**.

- In any **basic solution** the **basic variables** takes values of the form, $\sum_i \gamma_i b_i$, where the γ_i 's are either **0,1** or **-1**.
- In any **BFS x** of **TP** with supplies a_i , $i = 1, 2, \dots, m$ and demand d_j , $j = 1, 2, \dots, n$ the basic variables take values of the form,
 $x_{ij} = \sum_i \alpha_i a_i + \sum_j \beta_j d_j$ where the α_i 's, β_j 's are either **0,1** or **-1**.
- **Transportation Array (TA):** The mn variables x_{ij} can be arranged in an $m \times n$ array known as the $m \times n$ **transportation array**.
- In a transportation array (**TA**) each **cell** corresponds to a **variable**, that is the (i, j) th cell corresponds to variable x_{ij} .

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i .

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i .
- Similarly the n **columns** correspond to the n **demand constraints** and the sum of the values of the variables in **column** j is given by d_j .

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i .
- Similarly the n **columns** correspond to the n **demand constraints** and the sum of the values of the variables in **column** j is given by d_j .
- **Definition 1:** A subset of **cells** of the **TA** is said to be **linearly independent** if the set of **column vectors** in the matrix A corresponding to the cells (variables) are **linearly independent** (LI). Otherwise they are **linearly dependent** (LD).

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i .
- Similarly the n **columns** correspond to the n **demand constraints** and the sum of the values of the variables in **column** j is given by d_j .
- **Definition 1:** A subset of **cells** of the **TA** is said to be **linearly independent** if the set of **column vectors** in the matrix A corresponding to the cells (variables) are **linearly independent** (**LI**). Otherwise they are **linearly dependent** (**LD**).
- Any set of ≤ 2 cells is **LI**.

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i .
- Similarly the n **columns** correspond to the n **demand constraints** and the sum of the values of the variables in **column** j is given by d_j .
- **Definition 1:** A subset of **cells** of the **TA** is said to be **linearly independent** if the set of **column vectors** in the matrix A corresponding to the cells (variables) are **linearly independent** (**LI**). Otherwise they are **linearly dependent** (**LD**).
- Any set of ≤ 2 cells is **LI**.
- Any set of $\geq m + n$ cells is **LD**.

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i .
- Similarly the n **columns** correspond to the n **demand constraints** and the sum of the values of the variables in **column** j is given by d_j .
- **Definition 1:** A subset of **cells** of the **TA** is said to be **linearly independent** if the set of **column vectors** in the matrix A corresponding to the cells (variables) are **linearly independent** (**LI**). Otherwise they are **linearly dependent** (**LD**).
- Any set of ≤ 2 cells is **LI**.
- Any set of $\geq m + n$ cells is **LD**.
- **Definition 2:** A subset of $(m+n-1)$ **cells** of the **TA** is said to be a **basic set** of cells if they are **linearly independent**.

- The m **rows** correspond to the m **supply constraints**, hence the sum of the values of the variables in **row** i is given by a_i .
- Similarly the n **columns** correspond to the n **demand constraints** and the sum of the values of the variables in **column** j is given by d_j .
- **Definition 1:** A subset of **cells** of the **TA** is said to be **linearly independent** if the set of **column vectors** in the matrix A corresponding to the cells (variables) are **linearly independent** (**LI**). Otherwise they are **linearly dependent** (**LD**).
- Any set of ≤ 2 cells is **LI**.
- Any set of $\geq m + n$ cells is **LD**.
- **Definition 2:** A subset of $(m+n-1)$ **cells** of the **TA** is said to be a **basic set** of cells if they are **linearly independent**. The cells in a **basic set** are called **basic cells**.

- **Remark 2:** A **basic set** corresponds to a **basic solution** of the **TP**, where the variables corresponding to the **basic cells** are **basic variables** and the rest are **nonbasic variables**.

- **Remark 2:** A **basic set** corresponds to a **basic solution** of the **TP**, where the variables corresponding to the **basic cells** are **basic variables** and the rest are **nonbasic variables**.
- **Remark 3:** If $\mathcal{B}$ be a **basic set** of cells then the **submatrix** of $A_{(m+n-1) \times mn}$ with **columns** corresponding to $\mathcal{B}$, is a **basis matrix** **B**.

- **Remark 2:** A **basic set** corresponds to a **basic solution** of the **TP**, where the variables corresponding to the **basic cells** are **basic variables** and the rest are **nonbasic variables**.
- **Remark 3:** If $\mathcal{B}$ be a **basic set** of cells then the **submatrix** of $A_{(m+n-1) \times mn}$ with **columns** corresponding to $\mathcal{B}$, is a **basis matrix** **B**.
- There exists a **row** of **B** with exactly **one** nonzero entry.

- **Remark 2:** A **basic set** corresponds to a **basic solution** of the **TP**, where the variables corresponding to the **basic cells** are **basic variables** and the rest are **nonbasic variables**.
- **Remark 3:** If $\mathcal{B}$ be a **basic set** of cells then the **submatrix** of $A_{(m+n-1) \times mn}$ with **columns** corresponding to $\mathcal{B}$, is a **basis matrix** $\mathbf{B}$.
- There exists a **row** of $\mathbf{B}$ with exactly **one** nonzero entry.
- Since we are now solving $\mathbf{B}x_B = [a_1, \dots, a_m, d_1, \dots, d_{n-1}]^T$ and each row of $\mathbf{B}$ corresponds to a constraint (supply or demand).

- **Remark 2:** A **basic set** corresponds to a **basic solution** of the **TP**, where the variables corresponding to the **basic cells** are **basic variables** and the rest are **nonbasic variables**.
- **Remark 3:** If $\mathcal{B}$ be a **basic set** of cells then the **submatrix** of $A_{(m+n-1) \times mn}$ with **columns** corresponding to $\mathcal{B}$, is a **basis matrix** $\mathbf{B}$.
- There exists a **row** of $\mathbf{B}$ with exactly **one** nonzero entry.
- Since we are now solving $\mathbf{B}x_B = [a_1, \dots, a_m, d_1, \dots, d_{n-1}]^T$ and each row of $\mathbf{B}$ corresponds to a constraint (supply or demand).
- There exists a constraint in the **TP** which has **exactly one** basic variable (corresponding to $\mathbf{B}$).

- Since each **row** and **column** of the transportation array corresponds to a **constraint**, there exists a row or column of the transportation array which has **exactly one** cell from the basic set $\mathcal{B}$.

- Since each **row** and **column** of the transportation array corresponds to a **constraint**, there exists a row or column of the transportation array which has **exactly one** cell from the basic set $\mathcal{B}$.

If row i contains a single nonzero entry at the (i, j) th position, then the submatrix obtained from **B** after **deleting** the i th row and the j th column from **B** again has the same property.

- Since each **row** and **column** of the transportation array corresponds to a **constraint**, there exists a row or column of the transportation array which has **exactly one** cell from the basic set $\mathcal{B}$.

If row i contains a single nonzero entry at the (i, j) th position, then the submatrix obtained from **B** after **deleting** the i th row and the j th column from **B** again has the same property.

- If $\mathcal{B}$ is a **basic** set of cells and if the **row or column** having a single **basic cell** is struck off from the **TA**, then in the **reduced** (or remaining) **array** there is also a **row or column** with a **single basic cell**.

- Since each **row** and **column** of the transportation array corresponds to a **constraint**, there exists a row or column of the transportation array which has **exactly one** cell from the basic set $\mathcal{B}$.

If row i contains a single nonzero entry at the (i, j) th position, then the submatrix obtained from $\mathbf{B}$ after **deleting** the i th row and the j th column from $\mathbf{B}$ again has the same property.

- If $\mathcal{B}$ is a **basic** set of cells and if the **row or column** having a single **basic cell** is struck off from the **TA**, then in the **reduced** (or remaining) **array** there is also a **row or column** with a **single basic cell**.
- Since every row and column of the array has at least **one basic cell**, one can continue till **all** the rows and columns of the **TA** are **struck off**.

- Any set of $m + n - 1$ cells which satisfies the above property is a **basic set** of cells.

- Any set of $m + n - 1$ cells which satisfies the above property is a **basic set** of cells.
- **Example 1:** Consider the **TP** with a_i and d_j as given below:

- Any set of $m + n - 1$ cells which satisfies the above property is a **basic set** of cells.
- Example 1:** Consider the **TP** with a_i and d_j as given below:



	$j = 1$	2	3	4	5	6	a_i
$i = 1$							7
2							17
3							5
4							24
d_j	15	10	9	3	8	8	

- Any set of $m + n - 1$ cells which satisfies the above property is a **basic set** of cells.
- Example 1:** Consider the **TP** with a_i and d_j as given below:



	$j = 1$	2	3	4	5	6	a_i
$i = 1$							7
2							17
3							5
4							24
d_j	15	10	9	3	8	8	

- Let us first start with cell **(2,3)** as a basic cell and then try to construct a **BFS** of the above problem.

- Any set of $m + n - 1$ cells which satisfies the above property is a **basic set** of cells.
- Example 1:** Consider the **TP** with a_i and d_j as given below:



	$j = 1$	2	3	4	5	6	a_i
$i = 1$							7
2							17
3							5
4							24
d_j	15	10	9	3	8	8	

- Let us first start with cell (2,3) as a basic cell and then try to construct a **BFS** of the above problem.
- Since the minimum of a_2 and d_3 is $d_3 = 9$, we take $x_{23} = 9$. Delete the third column and change a_2 from 17 to $a'_2 = 17 - 9 = 8$.

- In the reduced array choose a basic cell say $(2, 4)$. Take $x_{24} = 3$ since $3 = \min\{d_4 = 3, a'_2 = 8\}$. Proceeding in this way we get the following **BFS**.

- In the reduced array choose a basic cell say $(2, 4)$. Take $x_{24} = 3$ since $3 = \min\{d_4 = 3, a'_2 = 8\}$. Proceeding in this way we get the following **BFS**.

	$j = 1$	2	3	4	5	6	a_j
$i = 1$		[7]					7
2			[9]	[3]	[5]		17
3					[3]	[2]	5
4	[15]	[3]				[6]	24
d_j	15	10	9	3	8	8	

North West Corner Rule for Initial BFS

North West Corner Rule for Initial BFS



					a_i
	40				40
					30
					60
d_j	50	20	40	20	

North West Corner Rule for Initial BFS



					a_i
	40				40
					30
					60
d_j	50	20	40	20	



					a_i
	40				40
	10				30
					60
d_j	50	20	40	20	

North West Corner Rule

North West Corner Rule

- Start by choosing a basic cell from the North-West corner of the array and assign the maximum possible value (maintaining feasibility) to that variable.



					a_i
	40				40
	10	20			30
					60
d_j	50	20	40	20	

North West Corner Rule

- Start by choosing a basic cell from the North-West corner of the array and assign the maximum possible value (maintaining feasibility) to that variable.



					a_i
	40				40
	10	20			30
					60
d_j	50	20	40	20	

- Delete the first row of the TA (the constraint being taken care of) and repeat the above process with the North-West corner of the reduced array.

North West Corner Rule



					a_i
	40				40
	10	20			30
		0			60
d_j	50	20	40	20	

North West Corner Rule



					a_i
	40				40
	10	20			30
		0			60
d_j	50	20	40	20	

- Delete the first column of the array and repeat the above process by starting the North-West corner of the reduced array.

North West Corner Rule



					a_i
	40				40
	10	20			30
		0			60
d_j	50	20	40	20	

- Delete the first column of the array and repeat the above process by starting the North-West corner of the reduced array. The following tables shows the various steps following the North-West corner rule till a **BFS** is obtained.

North West Corner Rule

North West Corner Rule



					a_i
	40				40
	10	20			30
		0	40		60
d_j	50	20	40	20	

North West Corner Rule

					a_i
	40				40
	10	20			30
		0	40		60
d_j	50	20	40	20	

					a_i
	40				40
	10	20			30
		0	40	20	60
d_j	50	20	40	20	

θ -loops

θ -loops

- A collection of cells of the **TA** is said to form a θ - loop if it satisfies the following conditions.

θ -loops

- A collection of cells of the **TA** is said to form a θ - loop if it satisfies the following conditions.
 1. Nonempty.

θ -loops

- A collection of cells of the **TA** is said to form a θ - loop if it satisfies the following conditions.
 1. Nonempty.
 2. Every row and column of the **TA** either has 0 or 2 cells from this collection.

θ -loops

- A collection of cells of the **TA** is said to form a θ - loop if it satisfies the following conditions.
 1. Nonempty.
 2. Every row and column of the **TA** either has 0 or 2 cells from this collection.
 3. No proper subset of this collection satisfies both property 1 and property 2.

- Consider the following examples.



	1	2	3	4
1	○	○		
2	○	○		
3				
4				

- Consider the following examples.



	1	2	3	4
1	○	○		
2	○	○		
3				
4				



	1	2	3	4
1	○			○
2	○	○		
3		○	○	
4			○	○



	1	2	3	4
1	☐			☐
2	☐	☐		
3		☐		
4				



	1	2	3	4
1	☐			☐
2	☐	☐		
3		☐		
4				



	1	2	3	4
1	☐	☐		
2	☐	☐		
3			☐	☐
4			☐	☐



	1	2	3	4
1	○			○
2	○	○		
3		○		
4				



	1	2	3	4
1	○	○		
2	○	○		
3			○	○
4			○	○

- In the third and fourth example, the marked cells do not form a θ -loop of the 4×4 transportation array, since it violates properties 2 and 3, respectively.



	1	2	3	4
1	○			○
2	○	○		
3		○		
4				



	1	2	3	4
1	○	○		
2	○	○		
3			○	○
4			○	○

- In the third and fourth example, the marked cells do not form a θ -loop of the 4×4 transportation array, since it violates properties 2 and 3, respectively.
- The first and the second however are θ -loops.

- **Theorem 4:** The cells in a θ - loop are **LD**.

- **Theorem 4:** The cells in a θ - loop are **LD**.
- **Corollary 5:** If a collection of cells contain a θ - loop then it is **LD**.

- **Theorem 4:** The cells in a θ - loop are **LD**.
- **Corollary 5:** If a collection of cells contain a θ - loop then it is **LD**.
- **Theorem 6 :** If $\triangle$ is a nonempty collection of cells of the **TA** which contains no θ - loop then it satisfies,

- **Theorem 4:** The cells in a θ - loop are **LD**.
- **Corollary 5:** If a collection of cells contain a θ - loop then it is **LD**.
- **Theorem 6 :** If $\triangle$ is a nonempty collection of cells of the **TA** which contains no θ - loop then it satisfies,
 1. There exists a row or column of the array with **exactly** one cell from $\triangle$.

- **Theorem 4:** The cells in a θ - loop are **LD**.
- **Corollary 5:** If a collection of cells contain a θ - loop then it is **LD**.
- **Theorem 6 :** If $\triangle$ is a nonempty collection of cells of the **TA** which contains no θ - loop then it satisfies,
 1. There exists a row or column of the array with **exactly** one cell from $\triangle$.
 2. Every nonempty subset of $\triangle$ should satisfy property 1.

- **Theorem 4:** The cells in a θ - loop are **LD**.
- **Corollary 5:** If a collection of cells contain a θ - loop then it is **LD**.
- **Theorem 6 :** If $\triangle$ is a nonempty collection of cells of the **TA** which contains no θ - loop then it satisfies,
 1. There exists a row or column of the array with **exactly** one cell from $\triangle$.
 2. Every nonempty subset of $\triangle$ should satisfy property 1.
- **Theorem 7 :** If $\triangle$ ($\neq \phi$) is a collection of cells of the **TA** which contains no θ - loop, then $\triangle$ is **LI**.

- **Theorem 4:** The cells in a θ - loop are **LD**.
- **Corollary 5:** If a collection of cells contain a θ - loop then it is **LD**.
- **Theorem 6 :** If $\triangle$ is a nonempty collection of cells of the **TA** which contains no θ - loop then it satisfies,
 1. There exists a row or column of the array with **exactly** one cell from $\triangle$.
 2. Every nonempty subset of $\triangle$ should satisfy property 1.
- **Theorem 7 :** If $\triangle (\neq \phi)$ is a collection of cells of the **TA** which contains no θ - loop, then $\triangle$ is **LI**.

Corollary 5 and **Theorem 7** implies the following:

- **Theorem 4:** The cells in a θ - loop are **LD**.
 - **Corollary 5:** If a collection of cells contain a θ - loop then it is **LD**.
 - **Theorem 6 :** If $\triangle$ is a nonempty collection of cells of the **TA** which contains no θ - loop then it satisfies,
 1. There exists a row or column of the array with **exactly** one cell from $\triangle$.
 2. Every nonempty subset of $\triangle$ should satisfy property 1.
 - **Theorem 7 :** If $\triangle (\neq \phi)$ is a collection of cells of the **TA** which contains no θ - loop, then $\triangle$ is **LI**.
- Corollary 5 and Theorem 7 implies the following:
- **Corollary 8:** A subset of cells $\triangle$ of the **TA** is **LI** if and only if it contains **no** θ - loop.

- **Theorem 9:** If $\mathcal{B}$ is a collection of $m + n - 1$ basic cells of the **TA** and $(p, q) \notin \mathcal{B}$, then $\mathcal{B} \cup \{(p, q)\}$ contains one and only one θ -loop and this loop includes the cell (p, q) .

- **Theorem 9:** If $\mathcal{B}$ is a collection of $m + n - 1$ basic cells of the TA and $(p, q) \notin \mathcal{B}$, then $\mathcal{B} \cup \{(p, q)\}$ contains one and only one θ -loop and this loop includes the cell (p, q) .
- **How to get the optimal solution from a given basic feasible solution?**

- **Theorem 9:** If $\mathcal{B}$ is a collection of $m + n - 1$ basic cells of the TA and $(p, q) \notin \mathcal{B}$, then $\mathcal{B} \cup \{(p, q)\}$ contains one and only one θ -loop and this loop includes the cell (p, q) .
- **How to get the optimal solution from a given basic feasible solution?**

Let $\mathbf{x} = (x_{ij})$ be the initial BFS.

- **Theorem 9:** If $\mathcal{B}$ is a collection of $m + n - 1$ basic cells of the **TA** and $(p, q) \notin \mathcal{B}$, then $\mathcal{B} \cup \{(p, q)\}$ contains one and only one θ -loop and this loop includes the cell (p, q) .
- **How to get the optimal solution from a given basic feasible solution?**

Let $\mathbf{x} = (x_{ij})$ be the initial BFS.

- The dual of the **TP** is given by

$$\text{Max} \sum_{i=1}^m a_i u_i + \sum_{j=1}^n d_j v_j$$

subject to,

$$u_i + v_j \leq c_{ij} \text{ for all } i = 1, \dots, m, j = 1, \dots, n.$$

- **Step 1:** For the basic cells corresponding to $\mathbf{x}$, $c_{ij} = u_i + v_j$.

- **Step 1:** For the basic cells corresponding to $\mathbf{x}$, $c_{ij} = u_i + v_j$.
- Solve this set of $m + n - 1$ equations for u_i and v_j .

- **Step 1:** For the basic cells corresponding to $\mathbf{x}$, $c_{ij} = u_i + v_j$.
- Solve this set of $m + n - 1$ equations for u_i and v_j .
- Since there are $m + n - 1$ equations and $m + n$, u_i , v_j 's, fix the value of any one variable and solve for the others.

- **Step 1:** For the basic cells corresponding to $\mathbf{x}$, $c_{ij} = u_i + v_j$.
- Solve this set of $m + n - 1$ equations for u_i and v_j .
- Since there are $m + n - 1$ equations and $m + n$, u_i, v_j 's, fix the value of any one variable and solve for the others.
- Since any one of the $(m + n)$ equations of the **TP** can be removed, one can take the corresponding variable of the dual say $v_n = 0$ and can consider that variable as absent from the equations $c_{ij} = u_i + v_j$.

- **Step 1:** For the basic cells corresponding to $\mathbf{x}$, $c_{ij} = u_i + v_j$.
- Solve this set of $m + n - 1$ equations for u_i and v_j .
- Since there are $m + n - 1$ equations and $m + n$, u_i , v_j 's, fix the value of any one variable and solve for the others.
- Since any one of the $(m + n)$ equations of the **TP** can be removed, one can take the corresponding variable of the dual say $v_n = 0$ and can consider that variable as absent from the equations $c_{ij} = u_i + v_j$.
- This set of equations is obtained from $\mathbf{y}^T \mathbf{B} = \mathbf{c}_B^T$, where $\mathbf{y}^T = [u_1, \dots, u_m, v_1, \dots, v_{n-1}]$.

- **Step 1:** For the basic cells corresponding to $\mathbf{x}$, $c_{ij} = u_i + v_j$.
- Solve this set of $m + n - 1$ equations for u_i and v_j .
- Since there are $m + n - 1$ equations and $m + n$, u_i , v_j 's, fix the value of any one variable and solve for the others.
- Since any one of the $(m + n)$ equations of the **TP** can be removed, one can take the corresponding variable of the dual say $v_n = 0$ and can consider that variable as absent from the equations $c_{ij} = u_i + v_j$.
- This set of equations is obtained from $\mathbf{y}^T \mathbf{B} = \mathbf{c}_B^T$, where $\mathbf{y}^T = [u_1, \dots, u_m, v_1, \dots, v_{n-1}]$.
- We have $m + n - 1$ equations and $m + n - 1$ unknowns, which can be easily solved.

- **Step 2:** Check if this **y** is feasible for the dual, that is if $u_i + v_j \leq c_{ij}$ for all the non basic cells. If yes, then stop.

- **Step 2:** Check if this **y** is feasible for the dual, that is if $u_i + v_j \leq c_{ij}$ for all the non basic cells. If yes, then stop.
- The corresponding BFS is then **optimal** for the primal.

- **Step 2:** Check if this **y** is feasible for the dual, that is if $u_i + v_j \leq c_{ij}$ for all the non basic cells. If yes, then stop.
- The corresponding BFS is then **optimal** for the primal.
- If not, then go to Step 3.

- **Step 2:** Check if this **y** is feasible for the dual, that is if $u_i + v_j \leq c_{ij}$ for all the non basic cells. If yes, then stop.
- The corresponding BFS is then **optimal** for the primal.
- If not, then go to Step 3.
- **Step 3:** Find the θ - loop in $B \cup \{(p, q)\}$, where the cell (p, q) is such that $c_{p,q} - u_p - v_q = \min\{c_{ij} - u_i - v_j : c_{ij} - u_i - v_j < 0\}$.

- **Step 2:** Check if this **y** is feasible for the dual, that is if $u_i + v_j \leq c_{ij}$ for all the non basic cells. If yes, then stop.
- The corresponding BFS is then **optimal** for the primal.
- If not, then go to Step 3.
- **Step 3:** Find the θ - loop in $B \cup \{(p, q)\}$, where the cell (p, q) is such that $c_{p,q} - u_p - v_q = \min\{c_{ij} - u_i - v_j : c_{ij} - u_i - v_j < 0\}$.
- **Step 4:** Assign value $+\theta$ to cell (p, q) and **alternately** assign $+\theta$ and $-\theta$ to all the cells in the θ - loop, so that the sum of the allocations ($+\theta$ and $-\theta$ allocations) in each row and column add up to zero.

- **Step 2:** Check if this **y** is feasible for the dual, that is if $u_i + v_j \leq c_{ij}$ for all the non basic cells. If yes, then stop.
- The corresponding BFS is then **optimal** for the primal.
- If not, then go to Step 3.
- **Step 3:** Find the θ - loop in $B \cup \{(p, q)\}$, where the cell (p, q) is such that $c_{p,q} - u_p - v_q = \min\{c_{ij} - u_i - v_j : c_{ij} - u_i - v_j < 0\}$.
- **Step 4:** Assign value $+\theta$ to cell (p, q) and **alternately** assign $+\theta$ and $-\theta$ to all the cells in the θ - loop, so that the sum of the allocations ($+\theta$ and $-\theta$ allocations) in each row and column add up to zero.
- Take $+\theta = \min\{x_{ij} \in \theta\text{-loop} : \text{cell } (i, j) \text{ is assigned value } -\theta\}$.

- **Step 2:** Check if this **y** is feasible for the dual, that is if $u_i + v_j \leq c_{ij}$ for all the non basic cells. If yes, then stop.
- The corresponding BFS is then **optimal** for the primal.
- If not, then go to Step 3.
- **Step 3:** Find the θ - loop in $\mathcal{B} \cup \{(p, q)\}$, where the cell (p, q) is such that $c_{p,q} - u_p - v_q = \min\{c_{ij} - u_i - v_j : c_{ij} - u_i - v_j < 0\}$.
- **Step 4:** Assign value $+\theta$ to cell (p, q) and **alternately** assign $+\theta$ and $-\theta$ to all the cells in the θ - loop, so that the sum of the allocations ($+\theta$ and $-\theta$ allocations) in each row and column add up to zero.
- Take $+\theta = \min\{x_{ij} \in \theta\text{-loop} : \text{cell } (i, j) \text{ is assigned value } -\theta\}$. Find the new BFS say **x'** where x'_{ij} is either equal to x_{ij} , $x_{ij} + \theta$ or $x_{ij} - \theta$.

- Now (p, q) is a basic cell.

- Now (p, q) is a basic cell.
- If $x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : (i, j) \text{ is assigned value } -\theta\}$, then the variable x_{rs} becomes a non basic variable in $\mathbf{x}'$.

- Now (p, q) is a basic cell.
- If $x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : (i, j) \text{ is assigned value } -\theta\}$, then the variable x_{rs} becomes a non basic variable in $\mathbf{x}'$.
- If there is a **tie** for **this minimum value**, choose any **one** as the leaving variable arbitrarily such that there are $(m + n - 1)$ basic cells in the next iteration.

- Now (p, q) is a basic cell.
- If $x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : (i, j) \text{ is assigned value } -\theta\}$, then the variable x_{rs} becomes a non basic variable in $\mathbf{x}'$.
- If there is a **tie** for **this minimum value**, choose any **one** as the leaving variable arbitrarily such that there are $(m + n - 1)$ basic cells in the next iteration.
- **Step 5:** Go to Step 1.

- Now (p, q) is a basic cell.
- If $x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : (i, j) \text{ is assigned value } -\theta\}$, then the variable x_{rs} becomes a non basic variable in $\mathbf{x}'$.
- If there is a **tie** for **this minimum value**, choose any **one** as the leaving variable arbitrarily such that there are $(m + n - 1)$ basic cells in the next iteration.
- **Step 5:** Go to Step 1.
- If x_{pq} is a **non basic variable** in a **BFS** then the corresponding column in the simplex table $B^{-1}\tilde{\mathbf{a}}_{p,q} = \mathbf{u}_{pq}$, is such that $u_{k,pq} = -1, 1$, or 0

- Now (p, q) is a basic cell.
- If $x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : (i, j) \text{ is assigned value } -\theta\}$, then the variable x_{rs} becomes a non basic variable in $\mathbf{x}'$.
- If there is a **tie** for **this minimum value**, choose any **one** as the leaving variable arbitrarily such that there are $(m + n - 1)$ basic cells in the next iteration.
- **Step 5:** Go to Step 1.
- If x_{pq} is a **non basic variable** in a **BFS** then the corresponding column in the simplex table $B^{-1}\tilde{\mathbf{a}}_{p,q} = \mathbf{u}_{pq}$, is such that $u_{k,pq} = -1, 1$, or 0 depending on whether the $\mathbf{k}$ th basic variable gets the allocation $\theta, -\theta$ or is not included in the θ -loop in $B \cup \{(p, q)\}$, respectively.

- If x_{pq} (or cell (p, q)) is the **entering variable** of the new basis then according to the minimum ratio rule of simplex, the **leaving variable** is (r, s) if

- If x_{pq} (or cell (p, q)) is the **entering variable** of the new basis then according to the minimum ratio rule of simplex, the **leaving variable** is (r, s) if
$$x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : \text{cell } (i, j) \text{ is assigned value } -\theta\}.$$

- If x_{pq} (or cell (p, q)) is the **entering variable** of the new basis then according to the minimum ratio rule of simplex, the **leaving variable** is (r, s) if

$$x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : \text{cell } (i, j) \text{ is assigned value } - \theta\}.$$
- **Example:** Consider the following transportation problem (P) with c_{ij} 's, a_i 's (40,30,30) and d_j 's (30,50,20) as given below:

2	5	1	40
1	4	5	30
1	5	3	30
30	50	20	

- If x_{pq} (or cell (p, q)) is the **entering variable** of the new basis then according to the minimum ratio rule of simplex, the **leaving variable** is (r, s) if $x_{rs} = \min\{x_{ij} \in \theta\text{-loop} : \text{cell } (i, j) \text{ is assigned value } - \theta\}$.
- **Example:** Consider the following transportation problem (P) with c_{ij} 's, a_i 's (40,30,30) and d_j 's (30,50,20) as given below:

2	5	1	40
1	4	5	30
1	5	3	30
30	50	20	

- Check whether the initial basic feasible solution $\mathbf{x}_0$ with basic cells $B = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 2)\}$, is optimal for (P) (by taking $v_2 = 0$, where v_2 is the dual variable corresponding to the second demand constraint). Also find the optimal solution.

- The BFS with $\mathcal{B} = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 2)\}$ as the basic cells is given by
 $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.

- The BFS with $\mathcal{B} = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 2)\}$ as the basic cells is given by $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.
- The other u_i, v_j values are obtained by solving the equations given by $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below:

- The BFS with $\mathcal{B} = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 2)\}$ as the basic cells is given by
 $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.
- The other u_i, v_j values are obtained by solving the equations given by $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below:
 - $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
 - $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$
 - $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
 - $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
 - $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.

- The BFS with $\mathcal{B} = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 2)\}$ as the basic cells is given by
 $x_{11} = 30, x_{12} = 10, x_{22} = 10, x_{23} = 20, x_{32} = 30$ as the values of the basic variables.
- The other u_i, v_j values are obtained by solving the equations given by $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below:
 - $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
 - $c_{12} - u_1 - v_2 = 0$, where $c_{12} = 5$
 - $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
 - $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
 - $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
- On solving by taking $v_2 = 0$ we get,
 $u_1 = 5, v_1 = -3, u_2 = 4, v_3 = 1, u_3 = 5$.

- The following table shows the $c_{ij} - u_i - v_j$ values against each cell.

- The following table shows the $c_{ij} - u_i - v_j$ values against each cell.
- Check that

- The following table shows the $c_{ij} - u_i - v_j$ values against each cell.
- Check that

$$c_{13} - u_1 - v_3 = 1 - 5 - 1 = -5, c_{21} - u_2 - v_1 = 1 - 4 - (-3) = 0,$$

$$c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1,$$

$$c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3,$$

- The following table shows the $c_{ij} - u_i - v_j$ values against each cell.
- Check that
 $c_{13} - u_1 - v_3 = 1 - 5 - 1 = -5$, $c_{21} - u_2 - v_1 = 1 - 4 - (-3) = 0$,
 $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:

0	0	-5	40
0	0	0	30
-1	0	-3	30
30	50	20	

- The following table shows the $c_{ij} - u_i - v_j$ values against each cell.
- Check that
 $c_{13} - u_1 - v_3 = 1 - 5 - 1 = -5$, $c_{21} - u_2 - v_1 = 1 - 4 - (-3) = 0$,
 $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:

0	0	-5	40
0	0	0	30
-1	0	-3	30
30	50	20	

- Since all the $c_{ij} - u_i - v_j$ values are not non negative, the above table is not optimal.

- The following table shows the $c_{ij} - u_i - v_j$ values against each cell.
- Check that
 $c_{13} - u_1 - v_3 = 1 - 5 - 1 = -5$, $c_{21} - u_2 - v_1 = 1 - 4 - (-3) = 0$,
 $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:

0	0	-5	40
0	0	0	30
-1	0	-3	30
30	50	20	

- Since all the $c_{ij} - u_i - v_j$ values are not non negative, the above table is not optimal.
- The most negative value of $c_{ij} - u_i - v_j$ is in cell (1, 3), so this will be the entering variable of the new BFS.

- The following table shows the $c_{ij} - u_i - v_j$ values against each cell.
- Check that
 $c_{13} - u_1 - v_3 = 1 - 5 - 1 = -5$, $c_{21} - u_2 - v_1 = 1 - 4 - (-3) = 0$,
 $c_{31} - u_3 - v_1 = 1 - 5 - (-3) = -1$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$, which is indicated in the following table:

	0	0	-5	40
	0	0	0	30
	-1	0	-3	30
	30	50	20	

- Since all the $c_{ij} - u_i - v_j$ values are not non negative, the above table is not optimal.
- The most negative value of $c_{ij} - u_i - v_j$ is in cell (1, 3), so this will be the entering variable of the new BFS.
- The unique θ - loop in $\mathcal{B} \cup \{(1, 3)\}$ is given by $\{(1, 2), (2, 2), (2, 3), (1, 3)\}$.

- Since (1, 3) is the entering variable, so if we give $+\theta$ allocation to cell (1, 3) (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.

- Since $(1, 3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1, 3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.

- Since $(1, 3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1, 3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 2)\}$ and the values of the basic variables are given by:
 $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30$.

- Since $(1, 3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1, 3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $B = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 2)\}$ and the values of the basic variables are given by:
 $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30$.
- If we take $u_1 = 0$, then solving for $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below,

- Since $(1, 3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1, 3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $B = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 2)\}$ and the values of the basic variables are given by:
 $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30$.
- If we take $u_1 = 0$, then solving for $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below,
- $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
 $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
 $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
 $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
 $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
 we get

- Since $(1, 3)$ is the entering variable, so if we give $+\theta$ allocation to cell $(1, 3)$ (or value of $x_{13} = +\theta$) then $x_{12} = 10 - \theta$, $x_{22} = 10 + \theta$, $x_{23} = 20 - \theta$.
- $x_{13} = 10$ is in the basis of the new BFS and x_{12} leaves the basis.
- New $B = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 2)\}$ and the values of the basic variables are given by:
 $x_{11} = 30, x_{13} = 10, x_{22} = 20, x_{23} = 10, x_{32} = 30$.
- If we take $u_1 = 0$, then solving for $c_{ij} - u_i - v_j = 0$ for the basic cells, that is by solving the 5 equations given below,
- $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
 $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
 $c_{23} - u_2 - v_3 = 0$, where $c_{23} = 5$
 $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
 $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
 we get
- $v_1 = 2, v_2 = 0, v_3 = 1, u_2 = 4, u_3 = 5$.

• $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$,
 $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$,
 $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
 $\mathcal{B} = \{(1, 1), (1, 3), (2, 3), (2, 2), (3, 2)\}$.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$,
 $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
 $B = \{(1, 1), (1, 3), (2, 3), (2, 2), (3, 2)\}$.

	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$,
 $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
 $B = \{(1, 1), (1, 3), (2, 3), (2, 2), (3, 2)\}$.

	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31} .

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$,
 $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
 $\mathcal{B} = \{(1, 1), (1, 3), (2, 3), (2, 2), (3, 2)\}$.



	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31} .
- The unique θ -loop in $\mathcal{B} \cup \{(3, 1)\}$ is given by $\{(3, 1), (3, 2), (2, 2), (2, 3), (1, 3), (1, 1)\}$.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$,
 $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
 $\mathcal{B} = \{(1, 1), (1, 3), (2, 3), (2, 2), (3, 2)\}$.



	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31} .
- The unique θ - loop in $\mathcal{B} \cup \{(3, 1)\}$ is given by $\{(3, 1), (3, 2), (2, 2), (2, 3), (1, 3), (1, 1)\}$.
- $(3, 1)$ is the entering variable, so if we give $+\theta$ allocation to cell $(3, 1)$ (or value of $x_{31} = +\theta$) then $x_{11} = 30 - \theta$,
 $x_{13} = 10 + \theta$, $x_{23} = 10 - \theta$, $x_{22} = 20 + \theta$, $x_{32} = 30 - \theta$.

- $c_{21} - u_2 - v_1 = 1 - 4 - 2 = -5$, $c_{12} - u_1 - v_2 = 5 - 0 - 0 = 5$,
 $c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$,
 $c_{33} - u_3 - v_3 = 3 - 5 - 1 = -3$.
- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with
 $\mathcal{B} = \{(1, 1), (1, 3), (2, 3), (2, 2), (3, 2)\}$.

	0	5	0	40
	-5	0	0	30
	-6	0	-3	30
	30	50	20	

- The entering variable for the new BFS is x_{31} .
- The unique θ - loop in $\mathcal{B} \cup \{(3, 1)\}$ is given by $\{(3, 1), (3, 2), (2, 2), (2, 3), (1, 3), (1, 1)\}$.
- $(3, 1)$ is the entering variable, so if we give $+\theta$ allocation to cell $(3, 1)$ (or value of $x_{31} = +\theta$) then $x_{11} = 30 - \theta$,
 $x_{13} = 10 + \theta$, $x_{23} = 10 - \theta$, $x_{22} = 20 + \theta$, $x_{32} = 30 - \theta$.
- So $\theta = 10$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $B = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $B = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.
- We take $u_1 = 0$, then by solving the 5 equations given below:

$$c_{11} - u_1 - v_1 = 0, \text{ where } c_{11} = 2$$

$$c_{13} - u_1 - v_3 = 0, \text{ where } c_{13} = 1$$

$$c_{22} - u_2 - v_2 = 0, \text{ where } c_{22} = 4$$

$$c_{31} - u_3 - v_1 = 0, \text{ where } c_{31} = 1$$

$$c_{32} - u_3 - v_2 = 0, \text{ where } c_{32} = 5.$$

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $B = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.
- We take $u_1 = 0$, then by solving the 5 equations given below:
 - $c_{11} - u_1 - v_1 = 0$, where $c_{11} = 2$
 - $c_{13} - u_1 - v_3 = 0$, where $c_{13} = 1$
 - $c_{22} - u_2 - v_2 = 0$, where $c_{22} = 4$
 - $c_{31} - u_3 - v_1 = 0$, where $c_{31} = 1$
 - $c_{32} - u_3 - v_2 = 0$, where $c_{32} = 5$.
- Check that $v_1 = 2, v_2 = 6, v_3 = 1, u_2 = -2, u_3 = -1$.

- The entering variable for the new BFS is $x_{31} = 10$ and x_{23} is the leaving variable.
- The values of the basic variables in the new BFS is given by $x_{11} = 20, x_{13} = 20, x_{22} = 30, x_{31} = 10, x_{32} = 20$.
- The basic set of cells is given by $B = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.
- We take $u_1 = 0$, then by solving the 5 equations given below:

$$c_{11} - u_1 - v_1 = 0, \text{ where } c_{11} = 2$$

$$c_{13} - u_1 - v_3 = 0, \text{ where } c_{13} = 1$$

$$c_{22} - u_2 - v_2 = 0, \text{ where } c_{22} = 4$$

$$c_{31} - u_3 - v_1 = 0, \text{ where } c_{31} = 1$$

$$c_{32} - u_3 - v_2 = 0, \text{ where } c_{32} = 5.$$
- Check that $v_1 = 2, v_2 = 6, v_3 = 1, u_2 = -2, u_3 = -1$.
- Check that $c_{23} - u_2 - v_3 = 5 - (-2) - 1 = 6, c_{21} - u_2 - v_1 = 1 - (-2) - 2 = 1, c_{12} - u_1 - v_2 = 5 - 0 - 6 = -1, c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

0	-1	0	40
1	0	6	30
0	0	3	30
30	50	20	

- The entering variable is x_{12} .

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12} .
- The θ -loop is $\{(3, 1), (3, 2), (1, 2), (1, 1)\}$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12} .
- The θ -loop is $\{(3, 1), (3, 2), (1, 2), (1, 1)\}$.
- If $x_{12} = +\theta$) then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12} .
- The θ -loop is $\{(3, 1), (3, 2), (1, 2), (1, 1)\}$.
- If $x_{12} = +\theta$) then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.
- Take $\theta = 20$.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12} .
- The θ -loop is $\{(3, 1), (3, 2), (1, 2), (1, 1)\}$.
- If $x_{12} = +\theta$ then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.
- Take $\theta = 20$.
Any one of x_{11} or x_{32} can be the leaving variable.

- The following table gives the $c_{ij} - u_i - v_j$ values for the above BFS with $\mathcal{B} = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}$.

	0	-1	0	40
	1	0	6	30
	0	0	3	30
	30	50	20	

- The entering variable is x_{12} .
- The θ -loop is $\{(3, 1), (3, 2), (1, 2), (1, 1)\}$.
- If $x_{12} = +\theta$ then $x_{11} = 20 - \theta$, $x_{31} = 10 + \theta$, $x_{32} = 20 - \theta$.
- Take $\theta = 20$.
Any one of x_{11} or x_{32} can be the leaving variable.
- Let x_{32} leave the basis.

- If we take $u_1 = 0$, then by solving the 5 equations given below:

$$c_{11} - u_1 - v_1 = 0, \text{ where } c_{11} = 2$$

$$c_{13} - u_1 - v_3 = 0, \text{ where } c_{13} = 1$$

$$c_{22} - u_2 - v_2 = 0, \text{ where } c_{22} = 4$$

$$c_{31} - u_3 - v_1 = 0, \text{ where } c_{31} = 1$$

$$c_{12} - u_1 - v_2 = 0, \text{ where } c_{12} = 5.$$

we get

- If we take $u_1 = 0$, then by solving the 5 equations given below:

$$c_{11} - u_1 - v_1 = 0, \text{ where } c_{11} = 2$$

$$c_{13} - u_1 - v_3 = 0, \text{ where } c_{13} = 1$$

$$c_{22} - u_2 - v_2 = 0, \text{ where } c_{22} = 4$$

$$c_{31} - u_3 - v_1 = 0, \text{ where } c_{31} = 1$$

$$c_{12} - u_1 - v_2 = 0, \text{ where } c_{12} = 5.$$

we get

- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1.$

- If we take $u_1 = 0$, then by solving the 5 equations given below:

$$c_{11} - u_1 - v_1 = 0, \text{ where } c_{11} = 2$$

$$c_{13} - u_1 - v_3 = 0, \text{ where } c_{13} = 1$$

$$c_{22} - u_2 - v_2 = 0, \text{ where } c_{22} = 4$$

$$c_{31} - u_3 - v_1 = 0, \text{ where } c_{31} = 1$$

$$c_{12} - u_1 - v_2 = 0, \text{ where } c_{12} = 5.$$

we get

- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1.$
- $c_{23} - u_2 - v_3 = 5 - (-1) - 1 = 5, c_{21} - u_2 - v_1 = 1 - (-1) - 2 = 0, c_{32} - u_3 - v_2 = 5 - (-1) - 5 = 1, c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3.$

- If we take $u_1 = 0$, then by solving the 5 equations given below:

$$c_{11} - u_1 - v_1 = 0, \text{ where } c_{11} = 2$$

$$c_{13} - u_1 - v_3 = 0, \text{ where } c_{13} = 1$$

$$c_{22} - u_2 - v_2 = 0, \text{ where } c_{22} = 4$$

$$c_{31} - u_3 - v_1 = 0, \text{ where } c_{31} = 1$$

$$c_{12} - u_1 - v_2 = 0, \text{ where } c_{12} = 5.$$

we get

- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1.$
- $c_{23} - u_2 - v_3 = 5 - (-1) - 1 = 5, c_{21} - u_2 - v_1 = 1 - (-1) - 2 = 0, c_{32} - u_3 - v_2 = 5 - (-1) - 5 = 1, c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3.$
- Since $c_{ij} - u_i - v_j \geq 0$ for all i, j ,
the above BFS is optimal and the optimal value is given by:

- If we take $u_1 = 0$, then by solving the 5 equations given below:

$$c_{11} - u_1 - v_1 = 0, \text{ where } c_{11} = 2$$

$$c_{13} - u_1 - v_3 = 0, \text{ where } c_{13} = 1$$

$$c_{22} - u_2 - v_2 = 0, \text{ where } c_{22} = 4$$

$$c_{31} - u_3 - v_1 = 0, \text{ where } c_{31} = 1$$

$$c_{12} - u_1 - v_2 = 0, \text{ where } c_{12} = 5.$$

we get

- $v_1 = 2, v_2 = 5, v_3 = 1, u_2 = -1, u_3 = -1.$
- $c_{23} - u_2 - v_3 = 5 - (-1) - 1 = 5, c_{21} - u_2 - v_1 = 1 - (-1) - 2 = 0, c_{32} - u_3 - v_2 = 5 - (-1) - 5 = 1, c_{33} - u_3 - v_3 = 3 - (-1) - 1 = 3.$
- Since $c_{ij} - u_i - v_j \geq 0$ for all i, j ,
the above BFS is optimal and the optimal value is given by:

$$c_{11}x_{11} + c_{12}x_{12} + c_{13}x_{13} + c_{22}x_{22} + c_{31}x_{31} =$$

$$2 \times 0 + 5 \times 20 + 1 \times 20 + 4 \times 30 + 1 \times 30 = 270.$$